Haozheng Pei

hpei8@gatech.edu | linkedin.com/in/haozheng-pei

Research interests: diffusion and flow matching, world models, generative-model evaluation and guarantees, probabilistic machine learning

EDUCATION

Georgia Institute of Technology Atlanta, GA

B.S. Mathematics; B.S. Industrial and Systems Engineering · Expected July 2027

Concentration: Artificial Intelligence and Operations Research; GPA: 3.77/4.00

Related coursework: Probability, Statistical Inference, Regression Analysis, Stochastic Processes, Machine Learning, Linear Algebra, PDE/ODE, Complex Analysis, Combinatorics, Operations Research

RESEARCH EXPERIENCE

Certified Mode-Recall for Generative Models May 2026 - Present

ICLR 2027 (under review); first author

- Developed a labeller-conditional, finite-sample framework to certify generative modes as present, absent, or inconclusive under imperfect classifier labels.
- Proved certificate validity and consistency, with sample-complexity bounds quantifying when rare modes can be certified.
- Evaluated GANs, flow matching, and pretrained DDPMs on 1,000-mode StackedMNIST, CIFAR-10, and 32 CelebA subgroups, using up to 40K generated samples per experiment.
- Obtained **valid bounds in 120/120 trials** and **zero false certifications over 20 seeds**; recovered **0.50 true coverage vs. 0.81 NN recall**, revealing a 0.97-to-0.50 subgroup-coverage drop missed by FID.

World Models for Robot Learning - RobotGen Studio Oct 2026 - Present

Research Assistant, Prof. Bo Zhu's group, Georgia Tech

- Contribute to a prototype project that aims to turn real-world images and videos into interactive simulation environments, robot tasks, demonstrations, and training data for vision-language-action models.
- Participate in literature review and research discussions on persistent memory and self-improvement through reuse of prior scenes, tasks, demonstrations, and execution experience.
- Explore demonstration generation combining code agents and programmable control with cloth, deformable-object, and fluid simulation.

Clink! Chop! Thud! Learning Object Sounds from Real-World Interactions Sep 2024 - Mar 2025

ICCV 2025; third author (of six); Georgia Tech Single Frame Segment Data Annotation Lab

- Contributed to a multimodal object-aware framework for sounding-object detection in in-the-wild egocentric video.
- Engineered annotation and feature pipelines that turned high-dimensional egocentric video into object-level supervision without manual labeling.
- Ran evaluation and ablations on EPIC-KITCHENS and Ego4D, where the method improved top-1 accuracy over the previous state of the art by 11.8 and 9.8 points.

RESEARCH EXPERIENCE (CONTINUED)

Probabilistic Forecasting for Wind Generation and Power Markets Aug 2025 - Aug 2026
Georgia Tech Electric Power Market Research

- Developed probabilistic forecasts of wind and solar generation using domain-knowledge-enhanced deep models (Transformer/DKFormer) alongside ARIMA and SVM baselines, improving accuracy and calibration relative to ISO benchmarks.
- Engineered spatiotemporal features from high-frequency turbine and weather data (10K+ assets); applied ensemble learning and Bayesian model combination to link forecast uncertainty to imbalance pricing, dispatch, and price formation.

SKILLS & LANGUAGES

Programming languages: Python, C++, Java, R

Libraries and tools: PyTorch, NumPy, SciPy, Pandas, Keras, scikit-learn, LightGBM, XGBoost, LaTeX, Git

Methods: Diffusion and flow matching, generative-model evaluation, finite-sample bounds, Monte Carlo simulation, probabilistic forecasting

INDUSTRY EXPERIENCE

UnionPay Advisors Shanghai, China
Data Science Intern | May 2026 - Aug 2026

- Built LightGBM credit-default models on 100K+ transactions, achieving **out-of-time AUC 0.78 and KS 0.46**.
- Engineered 700+ multi-horizon features; used WOE, chi-square binning, IV, and PSI/VIF filtering for stability (PSI < 0.001).

Hyde Park Investment Service New York, NY (Hybrid)
Quantitative Research Intern | Feb 2025 - Apr 2025

- Implemented an LLM-assisted sentiment pipeline on Wall Street Journal articles and combined it with macroeconomic data to forecast gold prices using logistic regression and XGBoost, achieving an AUC of approximately 0.8 and a backtested Sharpe ratio of 2.4.

TEACHING

Undergraduate Teaching Assistant, Georgia Tech Aug 2025 - Present
Teaching includes ISyE 3030, Basic Statistical Methods. Cumulative experience spans **five semesters and more than 100 students**, including preparation of teaching materials, tutoring and office hours, and grading assignments and examinations.

AWARDS

Center for Finance and Technology Scholar; AIME Qualified (11/15); AMC 12 Honor; High Distinction in the International Economics Olympiad.